Vibrations of Black Holes

In 1971 Bill Press, a student of mine at Caltech, discovered that black holes can vibrate at special, *resonant frequencies*, in much the same way as a violin string vibrates.

When a violin string is plucked just right, it emits a very pure tone: sound waves with a single frequency. When plucked a little differently, it emits that pure tone and also higher harmonics of the pure tone. In other words (if the string is firmly clamped, with the clamping finger not moving around) its vibrations produce sound at only a discrete set of frequencies, the string's resonant frequencies.

The same is true of a wine glass whose rim you rub with your finger, and a bell struck by a hammer. And also a black hole disturbed by something falling into it, Press discovered.

A year later Saul Teukolsky, another of my students, used Einstein's relativistic laws to work out a mathematical description of these resonant vibrations for a spinning black hole. (That's the best thing about teaching at Caltech; we get fabulous students!) By solving Teukolsky's equations, we physicists can compute a black hole's resonant frequencies. But solving them for an extremely fast spinning hole (like Gargantua) is very difficult. So difficult that it was not done successfully until forty years later—by a collaboration in which the lead players again were two Caltech students: Huan Yang and Aaron Zimmerman.

In September 2013, Ritchie Kremer, the property master for *Interstellar* (the person in charge of props) asked me for observational data that Romilly could show to Brand. Of course, I turned to the world's best experts for help: Yang and Zimmerman. They quickly produced tables of Gargantua's resonant vibration frequencies and also of the rates that the vibrations die out by feeding energy into